How does vehicle dominance change?

Centre for Blockchain Technologies

Vehicle Currency Rotation in Decentralised Exchanges

Evidence from Ethereum routes

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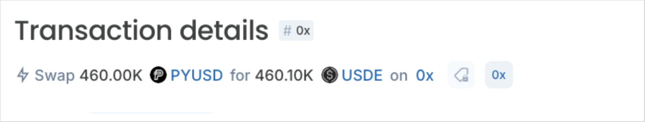
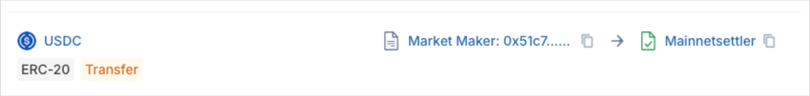
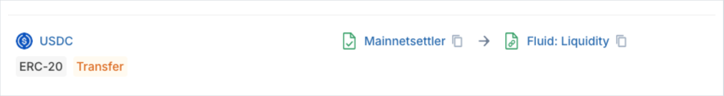
The PYUSD swap reaches a USDC-funded pool route

USER INSTRUCTION RECORDED BY THE EXPLORER



THE DECISIVE SETTLEMENT TRANSFERS





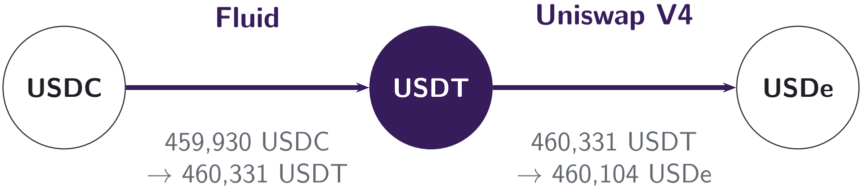
The instruction is PYUSD $\to$ USDe. The maker supplies the USDC used by the Fluid leg, which begins the connected pool route.

Note: The explorer screenshot and transfer trace refer to transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Inside the pools, USDT links USDC to USDe

ORDERED TRANSFERS AND ROUTE RECONSTRUCTION

User instruction: PYUSD → USDe ⟹ Pool route: USDC → USDT → USDe



vehicle

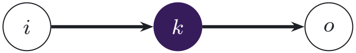
The observed ultimate trade is USDC → USDe. Its atomic trades are USDC → USDT and USDT → USDe; USDT is the vehicle that links them.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects ultimate pairs.

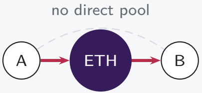


Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

Architecture changes routing opportunities

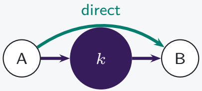
V1 mandatory hub



Only ETH–token pools: token-to-token execution must pass through ETH.

Hub use is imposed

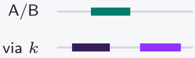
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

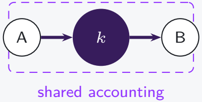
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



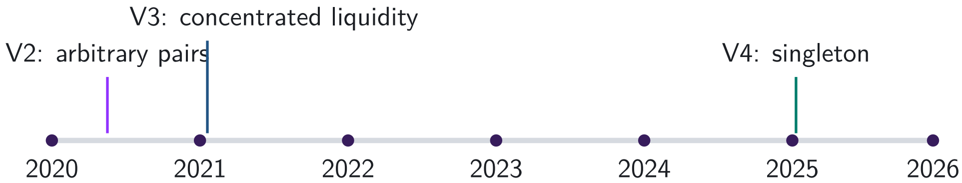
Two pools still execute; intermediate token movement may be netted.

Settlement can diverge from route use

Architecture feasible rules **Exposure** realised pool and route use **Calendar date** timing only

The route sample and architecture supplement answer different questions

DATA



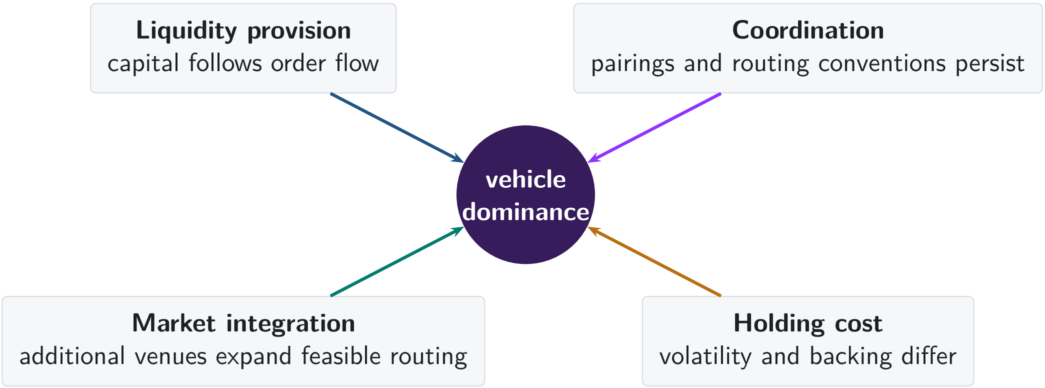
Main route sample

Atomic trades (swap legs) from multiple protocols are linked within each transaction. An atomic pair may be a two-asset pool or one asset pair inside a multi-asset Curve or Balancer pool.

V1/V2 architecture supplement

Uniswap V1 forced-ETH paths and V2 pair formation are analysed separately. They answer protocol-design questions and do not define the main route perimeter.

Why might one asset become the dominant vehicle?



Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every token pool pairs with ETH, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: MARKET CHOICE

95.5%

of single-leg V2 trades use a WETH pool in 2026

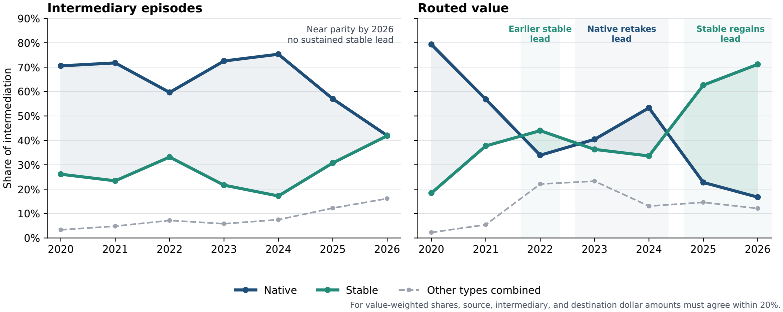
97.9%

of pairs first traded in 2026 include WETH

V2 converted ETH pairing from a protocol rule into a market choice. Native-asset concentration remained high through pair formation and realised trading.

Note: V1 counts linked token-to-ETH and ETH-to-token swaps with matching ETH quantities in one transaction. V2 shares use single-leg Uniswap V2 trades and newly traded pairs; they describe that venue rather than market-wide vehicle dominance.

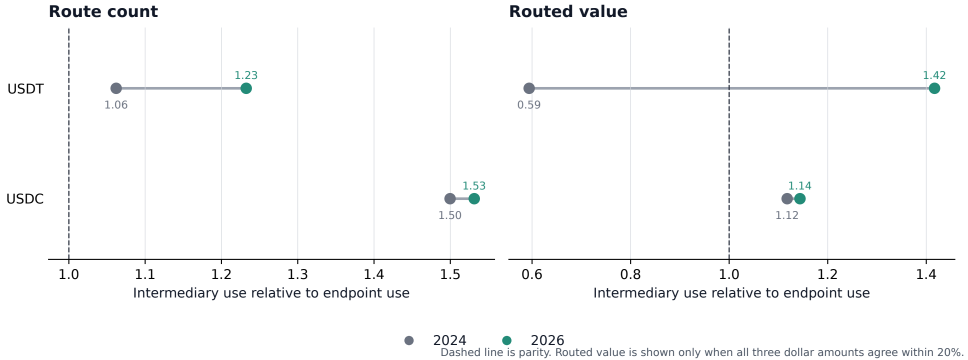
Stablecoins regain the lead in routed value



The aggregate share moves because the network of traded ultimate pairs reorganises around stablecoins. Holding the ultimate pair, the calendar date, and the route scope fixed, stablecoin share changes by $+0.2$ pp (SE 0.8 pp) by count and -1.3 pp (SE 2.2 pp) by value.

Note: The route universe is exact two-leg routes (two atomic trades) with an observed intermediary. The left panel counts one intermediary episode per route, hence per ultimate trade, and divides each asset type's episodes by all intermediary episodes. The right panel divides each type's routed value by total routed value among routes whose source, intermediary, and destination dollar amounts agree within 20%. The 2026 observation covers January–June; Other combines imported, liquid-staking, and residual assets. Matched-market estimates weight by route count or value and cluster by ordered ultimate pair and calendar date.

USDT drives the rise in stablecoin intermediation



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

Note: Excess use is intermediary share divided by endpoint share on the same route perimeter. The comparison uses native assets, USDC, and USDT and the same January–June dates in 2024 and 2026. Descriptive accounting: see Appendix A9; the identity carries no sampling interval.

The cross-section holds when the calendar is absorbed

	Native	Stable	Demand
Pooled	$+34.56$ pp	$+2.42$ pp	—
+ day effects	$+34.55$ pp	$+2.42$ pp	—
+ own demand	$+0.05$ pp	$+0.85$ pp	1.587
+ currency effects	—	—	0.985

Day effects leave the class premiums where they were. At the same trade demand the native premium is $+0.05$ pp (0.56 pp) and the stablecoin premium $+0.85$ pp (0.50 pp).

Note: 1,828,862 currency-days, 304,572 currencies, 2,259 days. Premiums are percentage points of the day's intermediary episodes against the residual unclassified bucket; demand is the pass-through of the currency's own route-endpoint share. Standard errors cluster on date and currency.

WETH-linked ultimate pairs account for most of the count rotation

WETH

WETH is the source or destination
WETH cannot also be the intermediary; stable share is mechanically 100%.

All matched trading-pair groups

$+21.1$ pp

95% CI [$+18.2$ pp, $+24.0$ pp]

WETH-linked route-count mass: 26.6% $\rightarrow$ 46.8%

$\longrightarrow$ exclude WETH endpoints

Trading-pair groups without WETH endpoints

$+3.7$ pp

95% CI [$+2.3$ pp, $+5.2$ pp]

79,347 matched groups

Within the same trading pair, calendar position, and route scope: all groups $+0.2$ pp [-1.3 pp, $+1.7$ pp]; without WETH endpoints $+0.3$ pp [-1.9 pp, $+2.6$ pp].

The count increase falls from $+21.1$ pp to $+3.7$ pp without WETH endpoints. A WETH-linked pair admits only a stablecoin intermediary in this comparison, so the count rotation largely records where trading happens rather than which intermediary a given trade selects.

Note: Exact two-leg native-WETH-versus-stablecoin routes on 181 common month-days. Intervals use 30-day Newey–West covariance on 362 endpoint-year daily observations; 2025 is excluded and actual dates preserve the gap. Within-group WLS fixes the ultimate pair, month-day, and route scope and clusters by pair and date. Feasible routes are not fixed; all comparisons are descriptive.

The routed-value rotation extends beyond WETH-linked ultimate pairs

Without WETH endpoints		Routed-value reallocation		Change within groups
$+21.5$ **pp**	=	$+23.5$ **pp**	+	-2.0 **pp**
95% CI $[+11.8$ pp, $+31.1$ pp$]$		95% CI $[+15.0$ pp, $+32.1$ pp$]$		95% CI $[-9.8$ pp, $+5.7$ pp$]$
strict-support stablecoin share of routed value Full matched sample: $+27.2$ pp $[+23.4$ pp, $+30.9$ pp$]$		Value shifts toward trading-pair and route-scope groups with higher stablecoin shares.		Stablecoin share changes within the same trading-pair and route-scope group.

Exact midpoint identity on each of 181 common calendar days; 2024 versus January–June 2026.

After removing WETH endpoints, value moves toward trading-pair and route-scope groups in which stablecoins already carry more intermediation. The change within those groups is small and imprecise.

Note: Routed value requires source, intermediary, and destination dollar amounts to agree within 20%. The daily midpoint identity is exact. Intervals use the same endpoint-year calendar-HAC covariance as the preceding frame; paired-month-day and bandwidth sensitivities remain in the source. Mukhin (2022), Gopinath and Stein (2021), and Carletti et al. (2021) guide the composition framing, not identification.

Three margins move an aggregate vehicle share

1 Used more often within a pair

USDC $\rightarrow$ WBTC trades in both years

Stablecoin share of its intermediary use

10.5% $\rightarrow$ 35.9%

Adds $+0.05$ pp to the aggregate

Margin total: -0.1 pp

2 Trading moves toward pairs that already use it

USDT $\rightarrow$ WETH, stablecoin-routed in both years

Share of all routed activity

0.36% $\rightarrow$ 4.52%

7,447 routes become 54,112; adds $+2.17$ pp

Margin total: $+8.6$ pp

3 Newly traded pairs route through it

USDS $\rightarrow$ WETH, absent from the base year

Routes observed

0 $\rightarrow$ 3,390

All stablecoin-intermediated; adds $+0.13$ pp

Margin total: $+21.0$ pp

The first margin is the one vehicle-currency theory describes, and it did not move. The third nets to $+17.8$ pp because 48.0% of routes run on pairs alive in only one year.

Note: Pooled route-count share among native and stablecoin intermediaries, 2024 against 2026 on 181 common calendar days. Each panel names the largest contributing ordered pair with two canonical symbols; unlabelled assets carry most of the third margin. Contributions are an exact allocation of the aggregate change and are descriptive.

Vehicle choice is sticky from market birth

5.0%

stable share at entry, 2024

25.1%

stable share at entry, 2026

$+97.0$ pp

30-day stable-born gap

Among 133,983 ordered pairs first observed in 2026, the first vehicle state predicts the next month: stable-born pairs stay at 97.5% stable share, while native-born pairs stay at 0.5%. At 120 days the gap remains $+98.8$ pp.

Note: Entrants are ordered ultimate pairs first observed in the matched January-through-June endpoint window with positive native-plus-stable primary-choice route mass. WETH-endpoint entrants carry 18.9% of 2026 entrant route mass; excluding them, entry stable share rises from 0.9% to 7.7%. Follow-up requires a complete 30-day window. Standard error for $+97.0$ pp clusters by calendar-day position. Descriptive association only.

Stable vehicles turn on at the market edge

-2.7 **pp**

stable turn-on per log baseline routes

$+27.1$ **pp**

stable turn-on per baseline direct-share unit

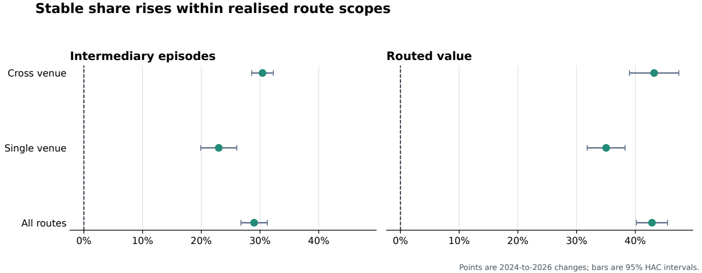
-1.1 **pp**

stable-majority switch per log baseline routes

In 94,260 transition rows, stable vehicles appear less often in already-large markets and more often where direct route availability is high. This is a driver screen, not a causal design.

Note: Count-share transition screen over 5,432 ordered pairs and 181 common month-days. Rows condition on month-day fixed effects and use harmonic endpoint-denominator weights; uncertainty is two-way CR1 by ordered pair and month-day. Complex-route baseline share is also positive for stable turn-on: $+20.3$ pp with SE 5.1 pp.

Stablecoin use rises within each venue scope



Neither venue mix nor path length absorbs the rotation: two-leg routes gain $+25.4$ pp (SE 1.05 pp), longer routes $+16.3$ pp (SE 1.86 pp), and the weakest venue-by-length cell still rises by $+17.0$ pp. Nor does the pricing rule or automated path search. Within a day the venue gap is a value phenomenon: cross-exchange one-intermediary routes carry $+14.18$ pp more stablecoin value share (SE 3.69 pp), against $+3.73$ pp on $[-2.2, +9.7]$ by count.

Note: Changes compare the same January–June dates in 2024 and 2026; bars show 95% calendar-HAC intervals. Excess-use ratios and the within-day venue gap span the full sample calendar.

Provider capital stays native-heavy

100.0%

WETH leads deposited capital

81.3%

stablecoins lead excess use

$4.2\times$

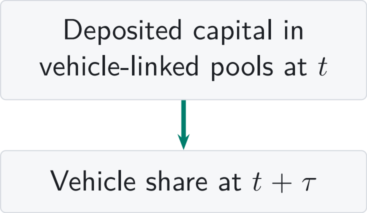
2026 route share over capital share

Stablecoins' candidate route share rises from 18.1% to 46.6%, while their candidate capital share falls from 17.6% to 11.1%. Within a date, after endpoint-share, venue-count, and pool-count controls, the stable label adds $+13.3$ pp to route share minus capital share.

Note: Daily five-candidate V2 panel over 2,239 route-and-capital-supported days. Capital is deposited-capital stock in candidate-linked V2 pools. Excess use is intermediary share divided by route-endpoint share. Leader rows are same-day rankings; level associations absorb candidate and origin-date effects. Descriptive association only.

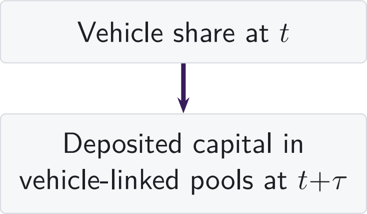
Deposited capital neither leads vehicle use nor follows it

LIQUIDITY FIRST



A one log-point higher capital stock predicts a next-day episode-share change of 0.00 pp (SE 0.04 pp); at thirty days -0.64 pp (SE 0.26 pp).

VEHICLE USE FIRST



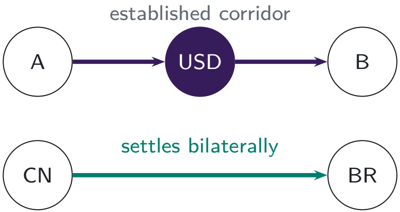
A one-point higher episode share predicts a next-day capital change of -0.03% (SE 0.03%); at thirty days -0.32% (SE 0.27%).

No use/capital measure pair is positive in both directions at any two of the one-, seven-, and thirty-day horizons; the smallest Holm-adjusted p is 0.17. The only strong longer-horizon pattern runs the other way: over the following 120 days, higher-capital pool-days lose episode share (-2.21 pp per log point, SE 0.39 pp).

Note: Daily panel of the five vehicle candidates on the full 2,238-day V2 calendar with candidate and date fixed effects; standard errors use 30-day calendar HAC on date-aggregated scores, and Holm adjustment is within direction; a month-block bootstrap and candidate-date clustering agree. Deposited capital is a reserve stock; provider flows and concentrated-liquidity depth are measured separately. Predictive associations only, not causal feedback.

The same arithmetic governs any aggregate currency share

A FAMILIAR SETTING



China and Brazil have built clearing arrangements for settling their bilateral trade in renminbi. The established corridor above is unaffected.

THE SHARE CAN FALL WITH NO CORRIDOR SWITCHING

A currency's share of all settled trade falls when trade grows fastest along corridors that never adopted it, even if every corridor that uses it today keeps using it tomorrow.

Margin 2 is trade reallocating toward corridors that already avoid the incumbent.

Margin 3 is a corridor that had no meaningful volume opening on other terms.

Our routes measure the same two margins directly, because the transaction log records the whole path rather than its legs.

Note: Illustrative comparison only. Renminbi clearing arrangements in Brazil follow the memorandum of understanding between the People's Bank of China and the Banco Central do Brasil. No currency share of trade settlement is measured here, and no causal or policy claim is made about either setting.

The market changed, not the trade

SHARE OF INTERMEDIATION CARRIED BY STABLECOINS

as the comparison holds more of the market fixed

Across the whole market		Within pairs traded in both years		Same pair, same day, same route scope
$+25.7$ **pp**	→	-0.1 **pp**	→	$+0.2$ **pp**
no conditioning at all		theory's own margin		SE 0.8 pp, over 94,260 matched cells

Stablecoins won through the markets that formed and grew, not repeated switching inside established pairs.

The three shares are one measure at three levels of conditioning and are directly comparable. They describe realised route composition; they are not entry or exit effects and do not identify a treatment.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

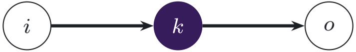
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Observed route endpoints

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

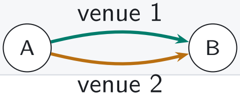
- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

How we reconstruct a route from one transaction

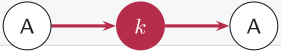
Direct swap
one transaction

Sequential vehicle route
one transaction

Parallel split
one transaction

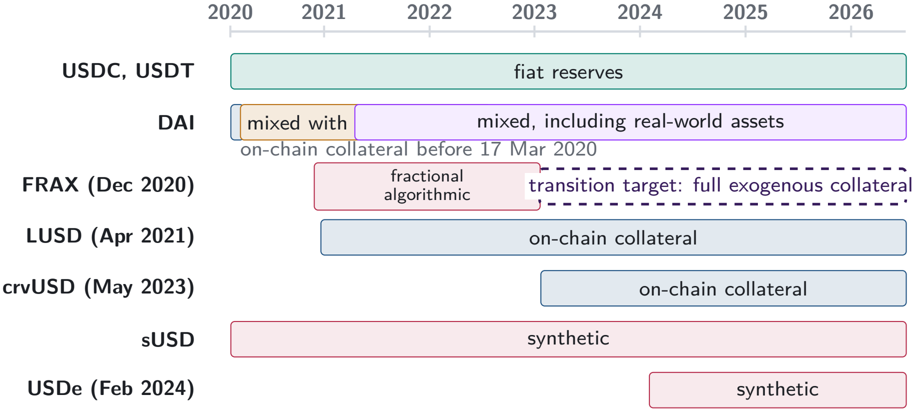


Round trip
one transaction



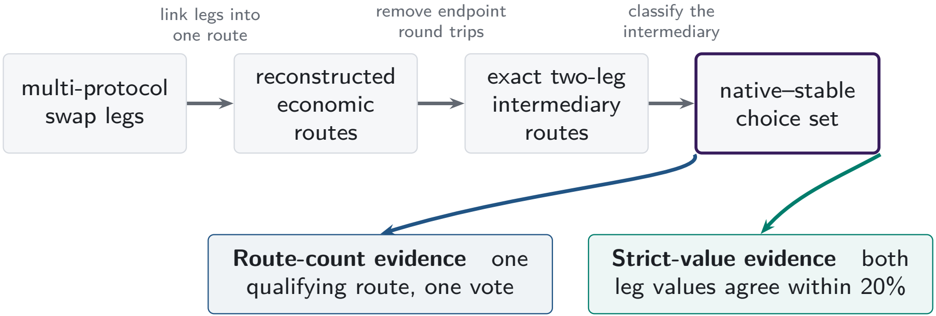
Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths reveal that choice rather than leaving it inside aggregate turnover.

A2. Stablecoin backing cannot be treated as fixed



Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

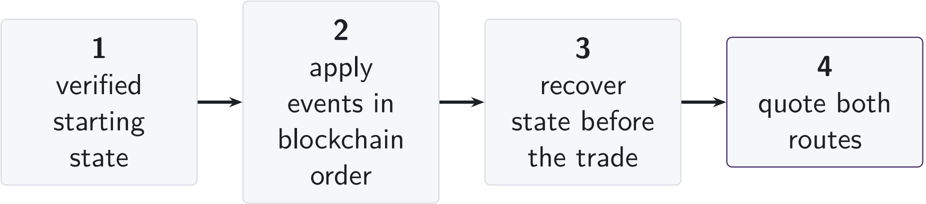
A3. One route universe supports two measurements



Box widths do not represent attrition.

The route sample links legs across protocols; one asset pair inside a multi-asset Curve or Balancer pool may supply a leg. The V1/V2 architecture evidence is separate. Dollar support affects value weighting, not route inclusion.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

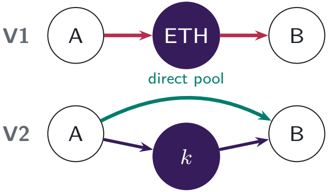
	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

The common object is the state immediately before execution. The fields required to recover that state vary with the pricing rule.

■ required state
□ hooks outside comparison

Protocol design changes markets, depth, and settlement

V1 $\longrightarrow$ V2: PAIR CREATION



V1 admits only ETH–token pools. V2 permits any ERC-20 pair, so market creation determines whether direct and vehicle paths coexist.

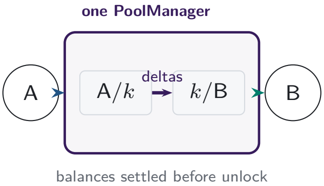
Which edges exist?

V3: ACTIVE LIQUIDITY

LPs choose a price range and fee tier for each pool. Positions covering the current price determine active liquidity in the direct pair and in each vehicle spoke.

Which path offers depth at this size?

V4: SHARED ACCOUNTING



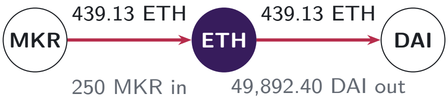
One PoolManager holds all pools. During a lock, the two swaps update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication. Pair creation changes available paths; concentrated liquidity changes pair-specific execution at the current state and trade size; the singleton changes the settlement boundary for a route that still uses two pools.

A6. V1 forced routes have an on-chain signature

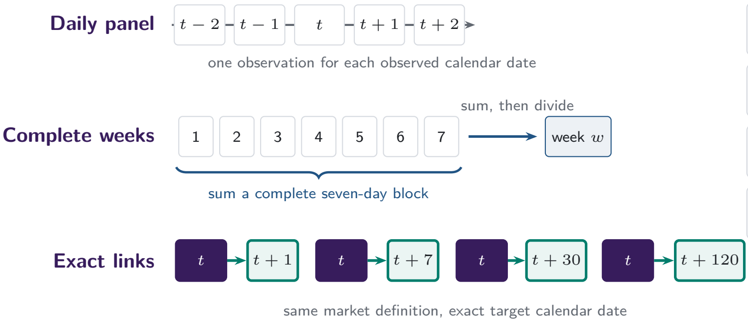
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction `0x4dca160d...96a0ca16` (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

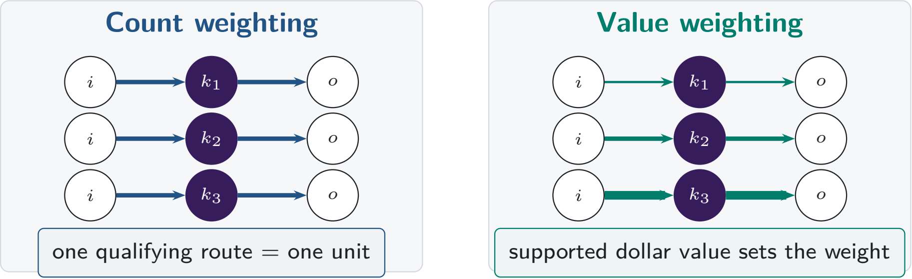
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

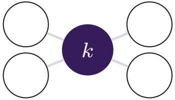
D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

Excess vehicle use

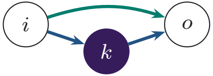
$X_k = D_k \div E_k$
E_k: endpoint share of k
$X_k > 1$: intermediary use exceeds the endpoint benchmark

Separate characteristics of the asset and route set



Network position

position of k in the feasible graph



Execution cost

which route returns more at the same state

Observed route-endpoint use supplies the benchmark; network position and same-state route costs remain separate characteristics.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I^N_{k,t} = \frac{n^I_{k,t}}{\sum_j n^I_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

$$E^N_{k,t} = \frac{n^E_{k,t}}{\sum_j n^E_{j,t}}$$

How often asset k appears at either end of the observed pool route.

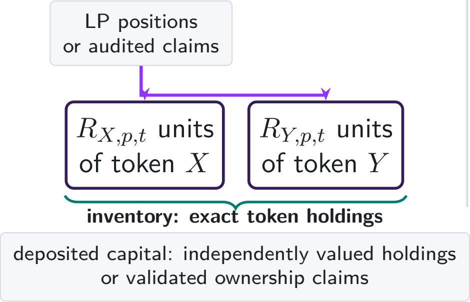
The benchmark describes reconstructed pool routes; an aggregator instruction can begin or end outside the connected component.

Excess use compares intermediation with observed route-endpoint use on the same support.

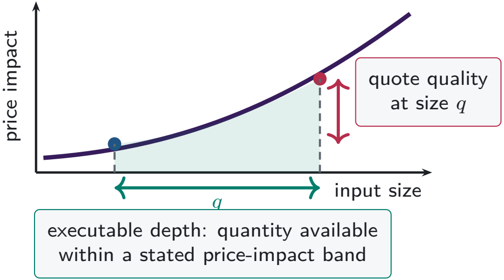
Note: For USDT in 2024–26, within-category changes account for 93.7% of the count change and 85.6% of the routed-value change; reweighting between single- and cross-venue categories accounts for 6.3% and 14.4%. The identity carries no sampling interval.

A10. Capital, inventory, and depth are different objects

Pool balance sheet at time t

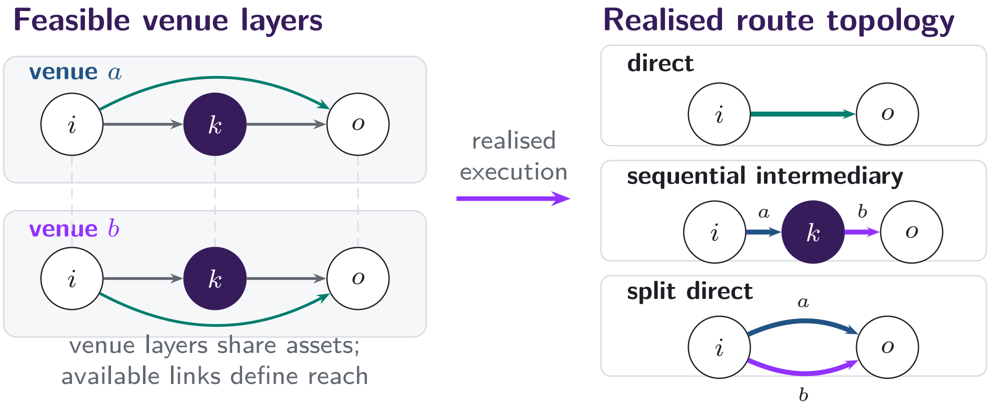


Execution from that state

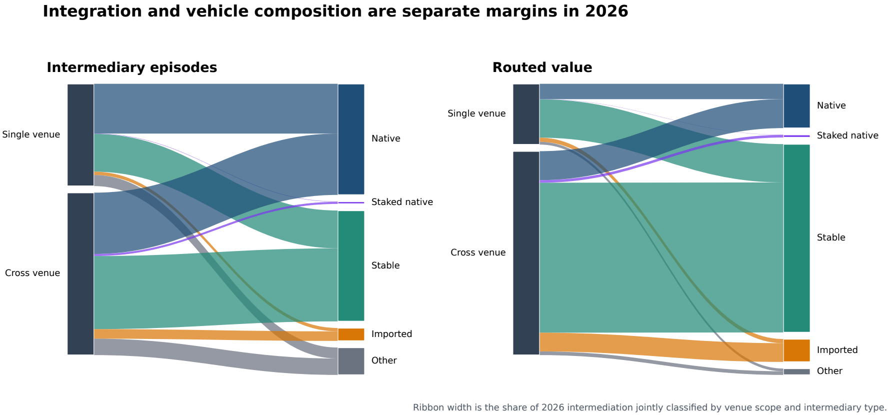


Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Additional venues expand the feasible route set



A11b. Venue scope and vehicle type differ in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage